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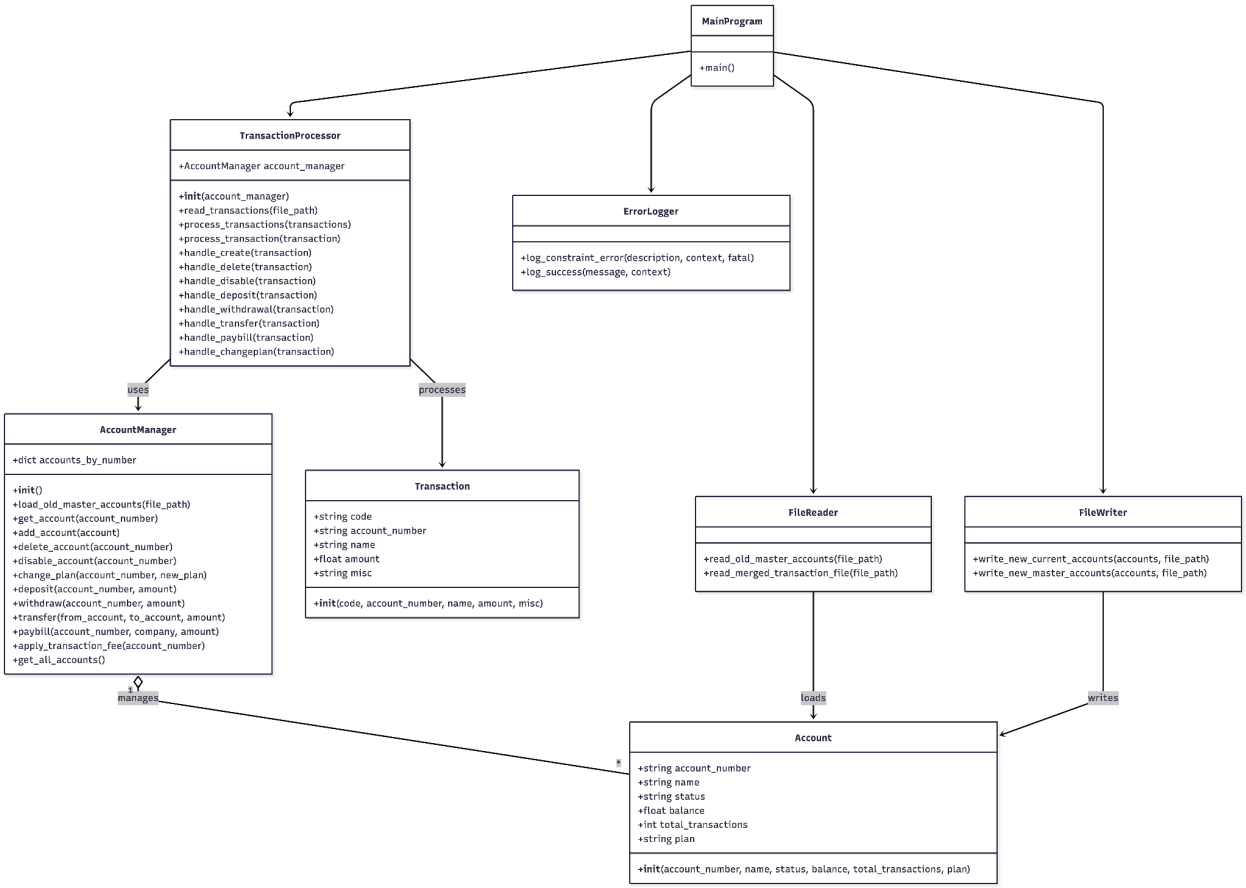
CSCI 3060U Course Project - Winter 2025

Phase #4 - Backend Rapid Prototype

UML Diagram:



Class and Method Intention Table:

Class / Method	Intention
MainProgram	Controls the overall execution of the Back End system.
main()	Starts the program, reads command-line arguments, and runs the full back end process.
TransactionProcessor	Processes merged transaction records and applies them to the loaded accounts.

init(account_manager)	Initializes the transaction processor with access to the account manager.
read_transactions(file_path)	Reads the merged transaction file and converts each record into a transaction object.
process_transactions(transactions)	Processes all transactions in order and applies valid changes to the accounts.
handle_create(transaction)	Determines the type of a transaction and sends it to the correct handler.
handle_delete(transaction)	Processes a delete-account transaction and removes an account if allowed.
handle_disable(transaction)	Processes a disable-account transaction and disables the selected account.
handle_deposit(transaction)	Processes a deposit transaction and adds money to an account.
handle_withdrawal(transaction)	Processes a withdrawal transaction and subtracts money if sufficient funds exist.
handle_transfer(transaction)	Processes a transfer transaction between two accounts.
handle_paybill(transaction)	Processes a paybill transaction and deducts the payment from the account.
handle_changeplan(transaction)	Processes a change-plan transaction and updates the account's plan type.
AccountManager	Stores and manages all bank accounts used by the Back End.
accounts_by_number	Stores all account objects using the account number as the key.
init()	Initializes the account manager and creates the account storage structure.
load_old_master_accounts(file_path)	Reads the old master accounts file and loads all accounts into memory.
get_account(account_number)	Returns the account associated with a given account number.
add_account(account)	Adds a newly created account to the system.

delete_account(account_number)	Deletes an account from the system when the deletion is valid.
disable_account(account_number)	Changes an account's status to disabled.
change_plan(account_number, new_plan)	Updates the plan type of an existing account.
deposit(account_number, amount)	Adds money to the balance of a specified account.
withdraw(account_number, amount)	Removes money from an account if the transaction is valid.
transfer(from_account, to_account, amount)	Transfers money from one account to another.
paybill(account_number, company, amount)	Deducts a bill payment amount from an account.
apply_transaction_fee(account_number)	Applies the appropriate transaction fee to an account.
get_all_accounts()	Returns all accounts so they can be written to the output files.
Transaction	Represents one transaction record from the merged daily transaction file.
code	Stores the transaction type code.
account_number	Stores the account number involved in the transaction.
name	Stores the account holder name associated with the transaction.
amount	Stores the amount involved in the transaction.
misc	Stores any additional transaction information needed for processing.
init(code, account_number, name, amount, misc)	Creates a transaction object using the fields read from the transaction file.
FileReader	Handles reading input files and converting them into data structures used by the system.
read_old_master_accounts(file_path)	Reads the old master accounts file and returns the loaded accounts.

read_merged_transaction_file(file_path)	Reads the merged transaction file and returns the loaded transactions.
FileWriter	Writes the updated account data into the required output file formats.
write_new_current_accounts(accounts, file_path)	Writes the new current accounts file in the required format.
write_new_master_accounts(accounts, file_path)	Writes the new master accounts file in the required format.
ErrorLogger	Handles all error reporting and logging for the system.
log_constraint_error(description, context, fatal)	Prints an error message and stops the program if the error is fatal.
log_success(message, context)	Prints a success message for a valid processed transaction.
Account	Represents one bank account and stores its data in memory.
account_number	Stores the unique 5-digit account number.
name	Stores the account holder's name.
status	Stores whether the account is active or disabled.
balance	Stores the current balance of the account.
total_transactions	Stores the number of successful transactions applied to the account.
plan	Stores the account plan type, such as student or non-student.
init(account_number, name, status, balance, total_transactions, plan)	Creates an account object with all required fields.

How to Run The Program:

python3 read.py master_accounts.txt temp_daily.txt stress_current.txt stress_master.txt

Example Output:

```
patrikprenga@Patriks-MacBook-Pro Phase4BackEnd % python3 read.py master_accounts.txt temp_daily.txt stress_current.txt stress_master.txt
ERROR: 05 ThisNameIsExactlyTwentyOneCharacters : Invalid amount format in transaction record
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12346
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12347
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12348
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12349
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12350
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12351
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12352
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12353
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12354
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12355
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12356
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12357
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12358
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12359
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12360
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12361
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12362
SUCCESS: 05 Jane Smith           00000 00500.00 0: Account created with new number 12363
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12364
SUCCESS: 05 EmptyAccount         00000 00000.00 0: Account created with new number 12365
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12366
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12367
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12368
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12369
ERROR: 01 User1                12345 00600.00 0: Withdrawal from invalid or disabled account
ERROR: 01 User1                12345 09999.00 0: Withdrawal from invalid or disabled account
ERROR: 01 Admin                12345 01000.00 0: Withdrawal from invalid or disabled account
ERROR: 01 User1                99999 00020.00 0: Withdrawal from invalid or disabled account
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12370
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12371
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12372
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12373
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12374
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12375
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12376
SUCCESS: 01 John Doe             00001 01000.00 0: Withdrawal processed. New balance = 50.00
SUCCESS: 01 John Doe             00001 00020.00 0: Withdrawal processed. New balance = 30.00
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12377
ERROR: 01 John Doe             00001 01000.00 0: Withdrawal would create a negative balance
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12378
ERROR: 01 John Doe             00001 01000.00 0: Withdrawal would create a negative balance
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12379
ERROR: 01 John Doe             00001 01000.00 0: Withdrawal would create a negative balance
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12380
ERROR: 01 John Doe             00001 01000.00 0: Withdrawal would create a negative balance
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12381
ERROR: 01 John Doe             00001 01000.00 0: Withdrawal would create a negative balance
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12382
ERROR: 01 John Doe             00001 01000.00 0: Withdrawal would create a negative balance
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12383
ERROR: 02 John Doe             00001 00100.00 0: Transfer would create a negative balance
ERROR: 01 John Doe             00001 01000.00 0: Withdrawal would create a negative balance
SUCCESS: 05 User1                00000 00200.00 0: Account created with new number 12384
ERROR: 02 John Doe             00001 00100.00 0: Transfer would create a negative balance
ERROR: 01 John Doe             00001 01000.00 0: Withdrawal would create a negative balance
```